TriPoseFusion: Robust Multi-View 3D Driver Pose Fusion with Outlier-Aware Canonicalization Supplementary Material

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1 Environment-Wise Results

Table S1 reports the environment-wise comparison that is summarized in the main paper. TriPoseFusion remains below both the raw best-single and canonicalized best-single baselines in all four illumination settings.

Table S1. Environment-wise comparison against raw and canonicalized best-single baselines. Values are averaged across 22 sequences per environment.

Environment	Best raw (m) ↓	Best canon. (m) ↓	TriPoseFusion (m) ↓	Δ vs canon. (m)
Day high	1.971	0.806	**0.435**	-0.371
Day low	2.284	1.133	**0.391**	-0.742
Night high	1.976	0.833	**0.418**	-0.415
Night low	2.081	0.936	**0.417**	-0.519

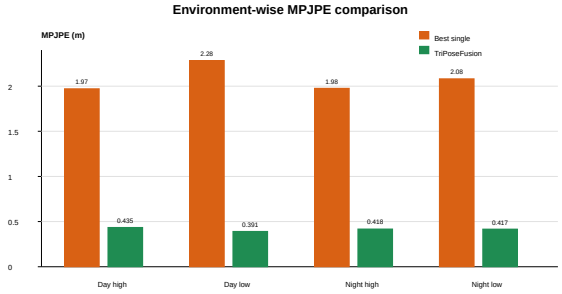


Fig. S1. Environment-wise MPJPE comparison. The plot visualizes the same environment-level trend as Table S1.

2 Additional Single-View Diagnostics

Table S2 reports a paired comparison between TriPoseFusion and an oracle best-single-view baseline over the 88 matched subject-environment pairs. The best-single baseline selects the lowest-error raw single camera for each sequence under the evaluation reference, so it is a diagnostic upper bound for fixed single-camera

selection rather than a deployable method. TriPoseFusion improves MPJPE and PCK@0.10 on every matched pair. The smaller PA-MPJPE win count reflects that Procrustes alignment can remove global similarity differences from single-view predictions, so PA-MPJPE should be interpreted together with MPJPE and PCK.

Table S2. Paired comparison against the raw oracle best-single-view baseline over 88 matched subject-environment pairs. Delta is TriPoseFusion minus best-single; lower is better for MPJPE/PA-MPJPE and higher is better for PCK.

Metric	Mean delta	TriPoseFusion better pairs
MPJPE (m) ↓	-1.224	88 / 88
PA-MPJPE (m) ↓	-0.426	29 / 88
PCK@0.10 ↑	+0.139	88 / 88

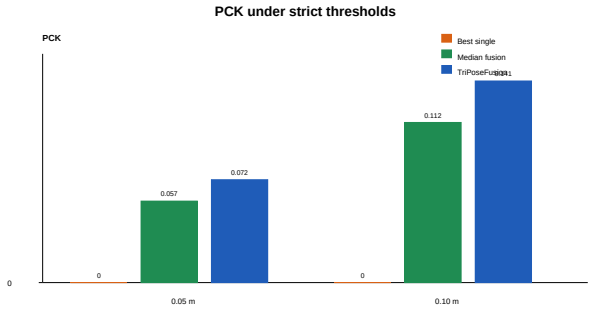


Fig. S2. PCK as a function of the distance threshold. This diagnostic complements the strict PCK@0.10 value reported in the main paper and shows the threshold sensitivity of the noisy upper-body and hand-focused pseudo-reference setting.

3 Diagnostic Ablations

Table S3 reports two-view diagnostic variants. These rows were kept out of the main paper because they are auxiliary diagnostics and the left+right variant has substantially different valid/skipped coverage. The results suggest that side views can be highly informative, but the coverage mismatch means the rows should not be read as direct replacements for the full three-view setting.

Table S3. Two-view diagnostic ablations. MPJPE and PA-MPJPE are reported in meters. Valid and Skipped denote evaluated or skipped temporal chunks.

Configuration	MPJPE (m) ↓	PA (m) ↓	PCK@0.10 ↑	Valid	Skipped
Full three-view model	0.412	0.275	0.141	21847	89
Full two-view front+left	0.411	0.273	0.141	21846	89
Full two-view front+right	0.415	0.279	0.141	21847	89
Full two-view left+right	**0.399**	**0.255**	**0.141**	20974	1506

Table S4 reports additional diagnostic ablations for attention, gate weighting, and temporal refinement. These rows are separated from the main paper because they come from updated diagnostic logs with different valid-chunk coverage.

Table S4. Additional diagnostic ablations. MPJPE and PA-MPJPE are reported in meters. Valid and Skipped denote evaluated or skipped temporal chunks.

Configuration	MPJPE (m) ↓	PA (m) ↓	PCK@0.10 ↑	Valid	Skipped
Full three-view model	0.412	0.275	0.141	21847	89
No cross-view attention	0.452	0.358	0.109	20424	88
Uniform gate	0.451	0.355	0.108	20424	88
Gate only, no TCN	0.452	0.358	0.108	20424	88

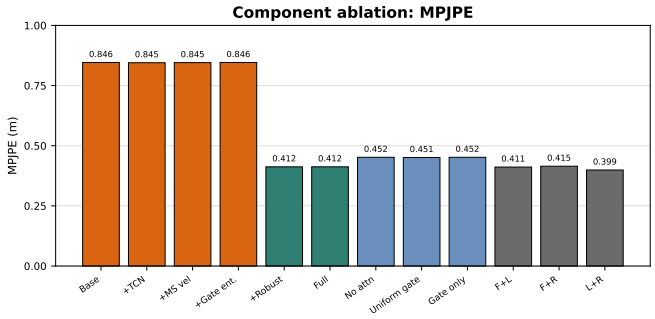


Fig. S3. Ablation MPJPE visualization. The largest improvement comes from robust body-centered canonicalization, while temporal and gating components provide smaller secondary effects under the reported pseudo-ground-truth protocol.

4 Gate-Weight Tables

Table S5 summarizes the joint-wise gate weights corresponding to the gate discussion in the main paper. Across the 52 joints, all three views remain within 5×10^{-5} of the uniform value 1/3, supporting the interpretation that the primary gain comes from robust canonicalization rather than strong average view selection. Table S6 reports the per-joint fusion gate values that were previously visualized as a figure.

Table S5. Summary statistics of learned joint-wise gate weights across the 52-keypoint model space. The final column reports the maximum absolute deviation from uniform weighting (1/3).

View	Mean	Std.	Min	Max	Max $\lvert w - 1/3 \rvert$
Front	0.333344	0.000019	0.333316	0.333382	0.000049
Left	0.333339	0.000020	0.333313	0.333378	0.000045
Right	0.333350	0.000017	0.333323	0.333379	0.000045

Table S6. Joint-wise fusion gate weights for the front, left, and right views. Names follow the SAM3D Body MHR70 metadata after applying the 52-keypoint subset used throughout the paper.

Joint	Name	Front	Left	Right	Joint	Name	Front	Left	Right
0	nose	0.333382	0.333377	0.333378	26	right_pinky_finger_third_joint	0.333334	0.333331	0.333351
1	left_eye	0.333381	0.333377	0.333379	27	right_wrist	0.333340	0.333339	0.333354
2	right_eye	0.333382	0.333377	0.333378	28	left_thumb4	0.333331	0.333325	0.333335
3	left_ear	0.333380	0.333378	0.333378	29	left_thumb3	0.333331	0.333324	0.333334
4	right_ear	0.333381	0.333378	0.333378	30	left_thumb2	0.333334	0.333327	0.333335
5	left_shoulder	0.333379	0.333378	0.333378	31	left_thumb_third_joint	0.333348	0.333340	0.333348
6	right_shoulder	0.333378	0.333378	0.333378	32	left_forefinger4	0.333342	0.333335	0.333349
7	right_thumb4	0.333317	0.333313	0.333324	33	left_forefinger3	0.333335	0.333328	0.333340
8	right_thumb3	0.333316	0.333313	0.333323	34	left_forefinger2	0.333330	0.333324	0.333333
9	right_thumb2	0.333318	0.333316	0.333326	35	left_forefinger_third_joint	0.333329	0.333323	0.333331
10	right_thumb_third_joint	0.333326	0.333324	0.333337	36	left_middle_finger4	0.333348	0.333335	0.333347
11	right_forefinger4	0.333322	0.333321	0.333336	37	left_middle_finger3	0.333348	0.333337	0.333350
12	right_forefinger3	0.333321	0.333320	0.333333	38	left_middle_finger2	0.333349	0.333341	0.333353
13	right_forefinger2	0.333319	0.333318	0.333331	39	left_middle_finger_third_joint	0.333340	0.333333	0.333341
14	right_forefinger_third_joint	0.333317	0.333315	0.333326	40	left_ring_finger4	0.333347	0.333333	0.333346
15	right_middle_finger4	0.333343	0.333338	0.333358	41	left_ring_finger3	0.333348	0.333334	0.333346
16	right_middle_finger3	0.333341	0.333338	0.333357	42	left_ring_finger2	0.333349	0.333337	0.333349
17	right_middle_finger2	0.333336	0.333336	0.333353	43	left_ring_finger_third_joint	0.333348	0.333338	0.333347
18	right_middle_finger_third_joint	0.333325	0.333325	0.333338	44	left_pinky_finger4	0.333342	0.333320	0.333338
19	right_ring_finger4	0.333340	0.333338	0.333357	45	left_pinky_finger3	0.333342	0.333323	0.333340
20	right_ring_finger3	0.333340	0.333339	0.333358	46	left_pinky_finger2	0.333344	0.333328	0.333343
21	right_ring_finger2	0.333340	0.333340	0.333359	47	left_pinky_finger_third_joint	0.333345	0.333330	0.333342
22	right_ring_finger_third_joint	0.333340	0.333340	0.333356	48	left_wrist	0.333344	0.333329	0.333340
23	right_pinky_finger4	0.333333	0.333328	0.333353	49	left_acromion	0.333379	0.333378	0.333378
24	right_pinky_finger3	0.333333	0.333329	0.333353	50	right_acromion	0.333379	0.333378	0.333378
25	right_pinky_finger2	0.333333	0.333330	0.333353	51	neck	0.333379	0.333378	0.333378